## Introduction

This report describes the stages of designing and implementing a moving display based on digital logic circuit principles (especially flip-flops), using the MultiSIM software environment.

The ultimate goal of this project is to build a matrix display composed of lamps that can show Persian or English letters and digits in a moving manner (from left to right or right to left).

The project was developed in three main phases, each building upon the achievements of the previous phase and gradually adding new capabilities to the circuit.

## Phase One: Design and Implementation of a Basic Universal Shift Register

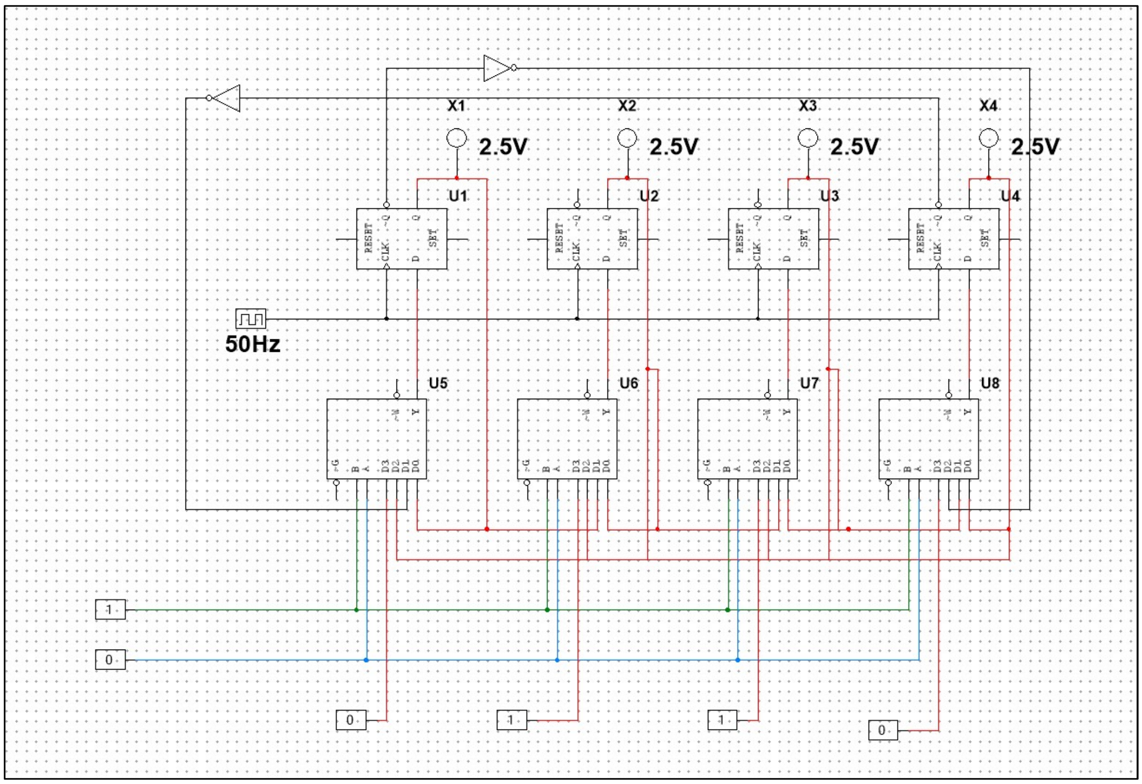


*Figure 1 – Phase One of the Project*

In the initial phase, the focus was on designing and implementing a 4-bit shift register circuit that serves as the core unit for generating and displaying a moving, repeating (circular) pattern.

This circuit consists of:

- Four D-type flip-flops
- Four 4-to-1 multiplexers
- Two NOT gates

Each flip-flop has:

- One data input
- One clock input
- Two outputs: Q and $\overline{Q}$ (bar-Q)

SET and RESET terminals are also available for initializing or clearing the flip-flop states. However, in this implementation, since no manual initialization or reset was required, these terminals were left unconnected.

The operation of the flip-flops is as follows:
At each active clock edge (rising edge in this design), the logical value at input D is transferred to output Q. Output $\overline{Q}$ is always the complement of Q.

All flip-flops are connected to a common clock source with a frequency of 50 Hz, which determines the shifting speed of the pattern and can be adjusted if needed. The shared clock ensures that all flip-flops update their states simultaneously, allowing synchronized and consistent data shifting throughout the register.

The universal capability of this shift register is achieved using four 4-to-1 multiplexers (U5 to U8), each connected to the input of one flip-flop.

Each multiplexer has:

- Four data inputs
- Two selection inputs labeled A and B

These two selection inputs determine which of the four data inputs is routed to the output of the multiplexer and subsequently to the D input of the corresponding flip-flop.

The control inputs A and B are connected in parallel across all multiplexers and are controlled using logic switches to define the operational mode of the entire register.

The operation based on control inputs A and B is as follows:

**Case 1**: A = 0, B = 0 (State "0")

The multiplexer selects input 0.
This input is connected to the Q output of the same flip-flop.

As a result, each flip-flop feeds back its current value to itself, causing the register contents to remain unchanged (hold mode).

**Case 2**: A = 0, B = 1 (State "1")

The multiplexer selects input 1.
For each flip-flop (except the first), this input is connected to the Q output of the flip-flop to its left (previous flip-flop).

This causes data to shift from left to right after each clock pulse.

For the first flip-flop, the output of the last flip-flop is used, making the shift operation circular.

Thus, the lamps move from left to right.

**Case 3**: A = 1, B = 0 (State “2”)

The multiplexer selects input 2.
For each flip-flop (except the last), this input is connected to the Q output of the flip-flop to its right (next flip-flop).

This causes data to shift from right to left after each clock pulse.

For the last flip-flop, the output of the first flip-flop is used to maintain circular operation.

Thus, the lamps move from right to left.

**Case 4**: A = 1, B = 1 (State “3”)

The multiplexer selects input 3.
This input is connected to an external data source.

This mode allows parallel loading of new data directly into each flip-flop.

In this way, the initial lamp pattern can be stored in the circuit:

- Lamps that should be ON are assigned value 1
- Lamps that should be OFF are assigned value 0

The NOT gates are used to neutralize the effect of $\overline{Q}$ outputs where necessary.

## Phase Two: Expansion of the Shift Register to 16 Bits

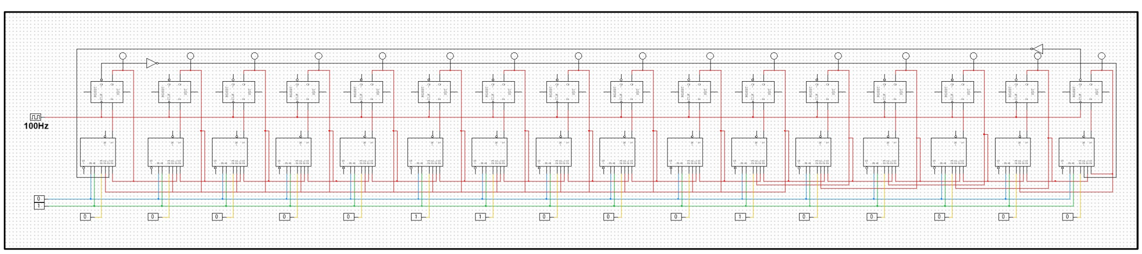


*Figure 2 – Phase Two of the Project*

In the second phase, since the display consists of 16 lamps, the shift register designed in Phase One was expanded.

The main objective was to increase the register capacity from 4 bits to 16 bits in order to allow more complex and longer patterns to be displayed across more output lamps.

The overall structure and operational principles remain the same as in Phase One, with the key difference being the increased number of flip-flops.

In this implementation:

- Sixteen D-type flip-flops are connected in series to form a 16-bit shift register.
- All 16 flip-flops share a common clock signal with a frequency of 100 Hz, which is adjustable.

The higher frequency compared to Phase One enables faster shifting across the 16 lamps.

The circuit ensures that once the pattern passes through all sixteen flip-flops, it returns to the first flip-flop, creating a continuous and repeating moving display across all 16 lamps.

## Phase Three (Final Phase): Implementation of Matrix Display and Motion Control

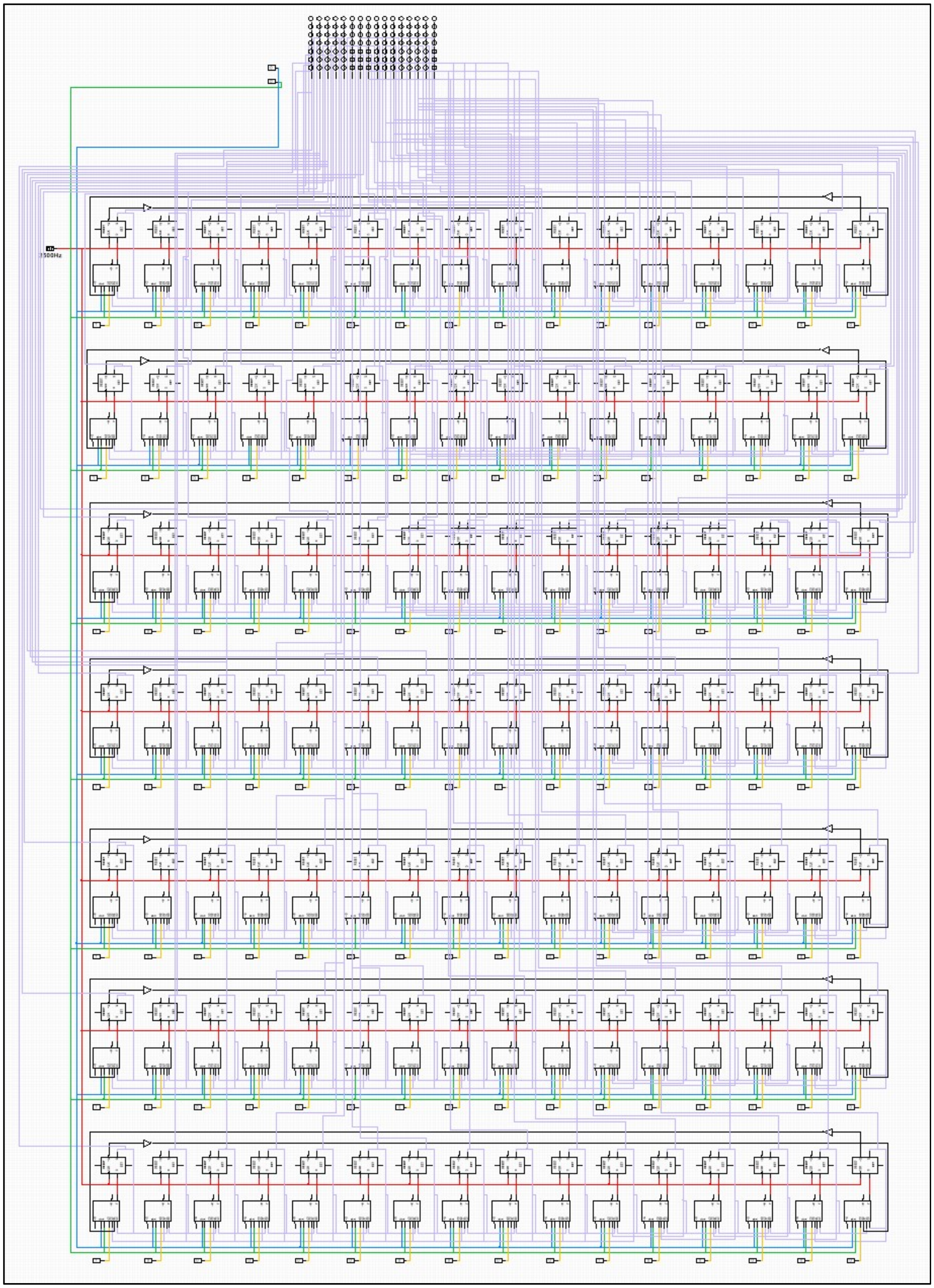


*Figure 3 – Final Phase of the Project*

In the final phase, the main goal was to extend the display capability from a linear (horizontal) format to a matrix (horizontal and vertical) format, enabling the display of more complex patterns such as letters and digits.

The display consists of:

- 7 rows
- Each row contains 16 lamps

Therefore, the 16-bit shift register designed in Phase Two is replicated 7 times, forming a 16 × 7 matrix structure.

For proper synchronization and smooth motion:

- All 42 flip-flops (7 rows × 16 flip-flops) are connected to a single common clock source.
- Due to the increased number of lamps and the need for smoother motion, the clock frequency is higher than in previous phases.
- The control switches of all multiplexers across the 7 rows are connected together so that any command (load, shift right, shift left) is applied simultaneously to all rows.

**Storing Custom Patterns**

One of the key capabilities in this phase is storing custom patterns in the lamp matrix.

This process is performed manually using the multiplexers:

1. First, the desired letter or digit pattern is defined on the 16 × 7 matrix.
2. Using control state "3", the lamps that should be ON are assigned value 1 and stored in the corresponding flip-flops.
3. Lamps that should remain OFF are assigned value 0.

After storing the pattern:

- If control state "1" is applied → pattern moves from left to right.
- If control state "2" is applied → pattern moves from right to left.
- If control state "0" is applied → pattern remains stationary and displays the stored data.

The main difference between state "0" and state "3" is:

- State "3" allows loading and modifying new data.
- State "0" only preserves and displays the previously stored pattern without modification.

**Output Connections**

The output of each flip-flop in every row is connected directly to its corresponding neon lamp in the display matrix.

Due to the size and complexity of the circuit (42 flip-flops and over 100 lamps), not all connections are clearly visible in the schematic.

For example:

- The output of the first flip-flop in the first row connects to lamp 1 in row 1.
- The second flip-flop in row 1 connects to lamp 2 in row 1.
- This continues sequentially until:
- The sixteenth flip-flop in row 7 connects to lamp 16 in row 7.

These precise connections ensure that each stored bit in the flip-flops is directly reflected in its corresponding lamp, enabling full and accurate control of each element in the matrix.